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1 Particle Decay and Rockets

1.1 a

Problem: The relative velocity v between two particles can be determined from the four-scalar product of their four-momenta,

$$P_1 \cdot P_2 = m_1 m_2 c^2 \gamma(v)$$

(where $\gamma(v) \equiv 1/\sqrt{1 - v^2/c^2}$). Show this formula and motivate well.

Solution: Let the first particle be in rest. Letting u_1 and u_2 be the four-velocities then

$$u_1 = \gamma(0)(c, \vec{0})$$

$$u_2 = \gamma(v)(c, \vec{v}).$$

Since $\gamma(0) = 1$ we can write the four-momentum as

$$P_1 = m_1 u_1 = m_1(c, \vec{0})$$

$$P_2 = m_2 u_2 = m_2 \gamma(v)(c, \vec{v}).$$

The calculation is therefore

$$P_1 \cdot P_2 = m_1 m_2 \gamma(v)(c, \vec{0}) \cdot (c, \vec{v}) = m_1 m_2 \gamma(v)(c^2 - \vec{0} \cdot \vec{v}) = m_1 m_2 c^2 \gamma(v)$$

□

1.2 b

Problem: Consider a particle A at rest, decaying into two particles B and C. Use suitable invariants to show that the velocity of B is determined by

$$\gamma(v_B) = \frac{m_A^2 + m_B^2 - m_C^2}{2m_A m_B}$$

Solution: In the center of momentum frame (CM-frame). Before the collision we have the four-momentum

$$P_A = (E_A/c, \vec{0})$$

where $E_A = m_A c^2$ and thus

$$P_A = m_A c(1, \vec{0})$$

Using the relation between four-momentum and four-velocity, after the collision we have the four-momenta

$$P_B = m_B u_B = m_B \gamma(v_B)(c, \vec{v}_B)$$

$$P_C = m_C u_C = m_C \gamma(v_C)(c, \vec{v}_C)$$

where u_B, u_C are the four-velocities and v_B, v_C are the three-velocities. Since the four-momentum is conserved we get that

$$P_A = P_B + P_C \iff P_C = P_A - P_B$$

Squaring both sides yields

$$\begin{aligned} P_C^2 &= P_A^2 + P_B^2 - 2P_A \cdot P_B \\ \iff m_C^2 c^2 &= m_A^2 c^2 + m_B^2 c^2 - 2P_A \cdot P_B \end{aligned}$$

Since A is stationary in the frame of reference, the relative velocity between A and B is

v_B . Thus we can use the result from a) which gives us

$$\begin{aligned} m_C^2 c^2 &= m_A^2 c^2 + m_B^2 c^2 - 2m_A m_B c^2 \gamma(v_B) \\ \iff 2m_A m_B \gamma(v_B) &= m_A^2 + m_B^2 - m_C^2 \\ \iff \gamma(v_B) &= \frac{m_A^2 + m_B^2 - m_C^2}{2m_A m_B} \end{aligned}$$

□

1.3 c

Problem: Imagine that we can fuel a rocket with an ideal fuel, all of whose mass is converted to energy, leaving only massless exhaust that is released straight backwards. A rocket, that initially is at rest, is packed with ideal fuel. The fuel makes up a fraction α of the loaded rocket's total mass. Derive the maximum velocity of the empty rocket after burning all the fuel, as a function of α . (*Answer:* $v = c \frac{1-(1-\alpha)^2}{1+(1-\alpha)^2}$. *Hint:* Consider the process as a “decay”.)

Solution: Let the rocket be stationary and of mass m before burning any fuel. Then after burning the mass of the rocket is $m(1 - \alpha)$. Writing the four-momenta before and after the burning gives

$$\begin{aligned} P_{\text{initial}} &= m(c, \vec{0}) \\ P_{\text{final}} &= m(1 - \alpha)(c, \vec{v}) \\ P_{\text{exhaust}} &= \vec{0} \end{aligned}$$

Assuming the burning of the fuel is instant, we can model the rocket as a decay reaction. Using the result from b) where A is the rocket before burning any fuel, B is the rocket

after burning fuel, and C is the massless exhaust, we obtain

$$\begin{aligned}
 \gamma(v) &= \frac{m^2 + m^2(1 - \alpha)^2 - 0}{2m^2(1 - \alpha)} \\
 \frac{1}{\sqrt{1 - v^2/c^2}} &= \frac{1 + (1 - \alpha)^2}{2(1 - \alpha)} \\
 \frac{2(1 - \alpha)}{1 + (1 - \alpha)^2} &= \sqrt{1 - v^2/c^2} \\
 \frac{4(1 - \alpha)^2}{1 + 2(1 - \alpha)^2 + (1 - \alpha)^4} &= 1 - v^2/c^2 \\
 v^2 &= c^2 - c^2 \frac{4(1 - \alpha)^2}{1 + 2(1 - \alpha)^2 + (1 - \alpha)^4} \\
 v^2 &= c^2 \left[\frac{1 + 2(1 - \alpha)^2 + (1 - \alpha)^4}{1 + 2(1 - \alpha)^2 + (1 - \alpha)^4} - \frac{4(1 - \alpha)^2}{1 + 2(1 - \alpha)^2 + (1 - \alpha)^4} \right] \\
 v^2 &= c^2 \frac{1 - 2(1 - \alpha)^2 + (1 - \alpha)^4}{1 + 2(1 - \alpha)^2 + (1 - \alpha)^4} = c^2 \frac{[1 - (1 - \alpha)^2]^2}{[1 + (1 - \alpha)^2]^2} \\
 v &= c \frac{1 - (1 - \alpha)^2}{1 + (1 - \alpha)^2}
 \end{aligned}$$

Answer: The maximum velocity as of the empty rocket as a function of α is

$$v(\alpha) = c \frac{1 - (1 - \alpha)^2}{1 + (1 - \alpha)^2}.$$